

SVKM'S NMIMS
MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

Academic Year: 2025-2026

Program: B.Tech Stream: Mechatronics

Year: 2nd Semester: IV

Subject: Dynamic Systems Modeling (702MH0C023)

Time: 45 Minutes

Date: 16-04-2026

No. of Pages: 1

Marks: 10

Set: 1

Test-II

Instructions: Candidates should read carefully the instructions.

1) Figures in brackets on the right hand side indicate full marks.

3) Assume Suitable data if necessary.

4) Question 1 is compulsory.

5) Answer any 2 from the remaining questions.

Q1

Answer briefly:

CO- 1,2;
SO- 1,2;
BL- 1,2

- a. (i) Define the Coriolis acceleration and state when it becomes significant in engineering analysis. [2]
(ii) What is an Atwood machine? State the underlying principle on which it operates.
(iii) Define the moment of inertia of a rigid body and state its SI unit.
(iv) Differentiate between undamped and damped free vibrations.
- b. (i) A solid disk of mass 5 kg and radius 0.2 m rolls without slipping. Calculate its moment of inertia about the contact point using the parallel axis theorem. [2]
(ii) Explain why a solid sphere reaches the bottom of an incline before a hollow cylinder of the same mass and radius.
(iii) In an Atwood machine with masses 5 kg and 3 kg, state which mass accelerates downward and why.
(iv) State two assumptions commonly made when analyzing an ideal pulley system.

Q2

CO- 2;
SO- 2;
BL- 3

- Two masses $m_1 = 5$ kg and $m_2 = 3$ kg are connected by a light, inextensible string passing over a massless, frictionless pulley. The system is released from rest. Take $g = 9.81$ m/s². [3]
- (a) Draw the free-body diagram for each mass and write the equation of motion for m_1 .
(b) Write the equation of motion for m_2 .
(c) Find the acceleration of the system.
(d) Calculate the tension in the string.
(e) Find the velocity of the masses after 2 seconds.
(f) Calculate the distance traveled by each mass in 2 seconds.

Q3

CO- 2;
SO- 2;
BL- 3,4

- A platform rotates at constant angular velocity $\Omega = 3$ rad/s about a vertical axis. An object at $r = 2$ m from the center moves radially outward at $v_{rel} = 1.5$ m/s with no radial acceleration ($\ddot{r} = 0$). The angular acceleration of the platform is zero ($\alpha = 0$). [3]
- (a) Calculate the centripetal acceleration magnitude.
(b) Calculate the Coriolis acceleration magnitude.
(c) State the direction of the Coriolis acceleration relative to the radial velocity.
(d) Find the total acceleration magnitude in the rotating frame.
(e) Determine the tangential velocity of the object as observed from the fixed (inertial) frame.
(f) If the object continues at the same radial speed, find the centripetal acceleration when $r = 4$ m.

Q4

CO- 2,3;
SO- 2,3;
BL- 4,5

- A uniform rod of mass $m = 2$ kg and length $L = 1$ m is pivoted at one end and oscillates as a physical pendulum under gravity. Take $g = 9.81$ m/s². [3]
- (a) Calculate the moment of inertia of the rod about the pivot using the parallel axis theorem.
(b) Find the distance of the center of mass from the pivot.
(c) Write the equation of motion for small-angle oscillations.
(d) Determine the natural frequency ω_n .
(e) Calculate the time period of oscillation.
(f) Compare this period with that of a simple pendulum of the same length ($L = 1$ m).

SVKM'S NMIMS
MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

Academic Year: 2025-2026

Program: B.Tech Stream: Mechatronics

Year: 2nd Semester: IV

Subject: Dynamic Systems Modeling (702MH0C023)

Time: 45 Minutes

Date: 16-04-2026

No. of Pages: 1

Marks: 10

Set: 2

Test-II

Instructions: Candidates should read carefully the instructions.

1) Figures in brackets on the right hand side indicate full marks.

3) Assume Suitable data if necessary.

4) Question 1 is compulsory.

5) Answer any 2 from the remaining questions.

Q1

Answer briefly:

CO- 1,2;
SO- 1,2;
BL- 1,2

- a. (i) State the parallel axis theorem and give its mathematical expression. [2]
(ii) Define the term 'gear ratio' and explain its physical significance.
(iii) What is a slider-crank mechanism? Name two practical applications.
(iv) Define critical damping and state the mathematical condition for it.

CO- 1,2;
SO- 1,2;
BL- 2,3

- b. (i) A gear train has an input speed of 100 RPM and a gear ratio of 15 : 1 (speed reduction). [2]
Calculate the output shaft speed.
(ii) Explain the physical meaning of the damping ratio ζ and what happens when $\zeta = 1$.
(iii) A CNC axis must traverse 200 mm with $v_{\max} = 100$ mm/s and $a_{\max} = 500$ mm/s².
Estimate the total traverse time using a trapezoidal profile.
(iv) State the condition for a block to remain stationary on an inclined plane with friction.

Q2

CO- 2;
SO- 2;
BL- 3

Two masses $m_1 = 8$ kg and $m_2 = 5$ kg are connected by a light, inextensible string passing over a massless, frictionless pulley. The system is released from rest. Take $g = 9.81$ m/s². [3]

- (a) Draw the free-body diagram for each mass and write the equation of motion for m_1 .
(b) Write the equation of motion for m_2 .
(c) Find the acceleration of the system.
(d) Calculate the tension in the string.
(e) Find the velocity of the masses after 3 seconds.
(f) Calculate the distance traveled by each mass in 3 seconds.

Q3

CO- 2;
SO- 2;
BL- 3,4

A block of mass $m = 5$ kg is placed on a rough inclined plane at angle $\theta = 30^\circ$ to the horizontal. [3]
The coefficient of kinetic friction is $\mu_k = 0.2$. The block is released from rest. Take $g = 9.81$ m/s².

- (a) Draw the free-body diagram showing all forces on the block.
(b) Calculate the normal reaction force N .
(c) Calculate the kinetic friction force f .
(d) Determine the net force along the incline and find the acceleration.
(e) Find the velocity of the block after sliding 5 m from rest.
(f) Calculate the time taken to slide 5 m from rest.

Q4

CO- 2,3;
SO- 2,3;
BL- 4,5

A mass $m = 2$ kg is attached to a spring of stiffness $k = 200$ N/m. The system is displaced $x_0 = 0.1$ m from equilibrium and released from rest ($v_0 = 0$). [3]

- (a) Write the equation of motion for the undamped system.
(b) Calculate the natural frequency ω_n and the time period T .
(c) Write the displacement as a function of time $x(t)$ for the given initial conditions.
(d) A viscous damper with $c = 4$ N·s/m is now added. Calculate the critical damping coefficient c_{cr} and the damping ratio ζ .
(e) State whether the damped system is underdamped, critically damped, or overdamped.
Calculate the damped natural frequency ω_d .
(f) Calculate the logarithmic decrement δ for the damped system.

SVKM'S NMIMS
MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

Academic Year: 2025-2026

Program: B.Tech Stream: Mechatronics

Year: 2nd Semester: IV

Subject: Dynamic Systems Modeling (702MH0C023)

Time: 45 Minutes

Date: 16-04-2026

No. of Pages: 1

Marks: 10

Set: 3

Test-II

Instructions: Candidates should read carefully the instructions.

- 1) Figures in brackets on the right hand side indicate full marks.
- 3) Assume Suitable data if necessary.
- 4) Question 1 is compulsory.
- 5) Answer any 2 from the remaining questions.

Q1

Answer briefly:

CO- 1,2;
SO- 1,2;
BL- 1,2

- a. (i) Explain the concept of rolling without slipping and state the velocity constraint. [2]
(ii) What is a trapezoidal velocity profile in CNC motion planning? Sketch it qualitatively.
(iii) Define logarithmic decrement in vibration analysis and state its physical meaning.
(iv) State the relationship between mechanical advantage and the number of supporting rope segments in a block-and-tackle system.
- b. (i) A platform rotates at $\Omega = 3$ rad/s. A particle moves radially outward at $v_{rel} = 2$ m/s. Calculate [2]
the magnitude of the Coriolis acceleration.
(ii) Write the equation of motion for an undamped spring-mass system and identify each term.
(iii) Define mechanical advantage of a pulley system. State its ideal value for a single movable pulley.
(iv) Explain why fictitious forces (pseudo-forces) appear in a rotating (non-inertial) reference frame.

Q2

CO- 2;
SO- 2;
BL- 3

- Two masses $m_1 = 6$ kg and $m_2 = 4$ kg are connected by a light, inextensible string passing over a [3]
massless, frictionless pulley. The system is released from rest. Take $g = 9.81$ m/s².
(a) Draw the free-body diagram for each mass and write the equation of motion for m_1 .
(b) Write the equation of motion for m_2 .
(c) Find the acceleration of the system.
(d) Calculate the tension in the string.
(e) Find the velocity of the masses after 2 seconds.
(f) Calculate the distance traveled by each mass in 2 seconds.

Q3

CO- 2;
SO- 2;
BL- 3,4

- A slider-crank mechanism has crank radius $r = 50$ mm, connecting rod length $L = 150$ mm, and [3]
operates at 3000 RPM. Analyze the mechanism at crank angle $\theta = 30^\circ$.
(a) Convert the crank speed from RPM to rad/s.
(b) Calculate the connecting rod angle ϕ using the kinematic relation $r \sin \theta = L \sin \phi$.
(c) Determine the piston displacement from top dead center (TDC).
(d) Find the piston velocity.
(e) Identify the crank angles at which the piston velocity is zero.
(f) State the type of motion executed by the piston and name one engineering application.

Q4

CO- 2,3;
SO- 2,3;
BL- 4,5

- A yo-yo of mass $m = 0.05$ kg has outer radius $R = 0.03$ m, axle radius $r = 0.01$ m, and moment of [3]
inertia $I = 2.25 \times 10^{-5}$ kg·m² (modeled as a solid disk of radius R). The string unwinds from the
axle. Take $g = 9.81$ m/s².
(a) Write the translational equation of motion for the yo-yo (vertical forces).
(b) Write the rotational equation of motion about the center of mass.
(c) State the constraint relating linear acceleration a and angular acceleration α .
(d) Solve for the linear acceleration of the yo-yo.
(e) Calculate the tension in the string.
(f) Express the acceleration as a fraction of g and explain why the yo-yo descends slower than free
fall.

SVKM'S NMIMS
MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

Academic Year: 2025-2026

Program: B.Tech Stream: Mechatronics

Year: 2nd Semester: IV

Subject: Dynamic Systems Modeling (702MH0C023)

Time: 45 Minutes

Date: 16-04-2026

No. of Pages: 1

Marks: 10

Set: 4

Test-II

Instructions: Candidates should read carefully the instructions.

1) Figures in brackets on the right hand side indicate full marks.

3) Assume Suitable data if necessary.

4) Question 1 is compulsory.

5) Answer any 2 from the remaining questions.

Q1

Answer briefly:

CO- 1,2;
SO- 1,2;
BL- 1,2

- a. (i) Define a holonomic constraint and give one example from a mechanical system. [2]
(ii) What is the physical significance of the natural frequency of a vibrating system?
(iii) Define the centripetal and Coriolis components of acceleration in a rotating reference frame.
(iv) What is a physical (compound) pendulum? How does it differ from a simple pendulum?
- b. (i) A slider-crank has crank radius $r = 50$ mm and connecting rod length $L = 150$ mm. State the ratio L/r and explain its significance in mechanism design. [2]
(ii) For a spring-mass system with $m = 2$ kg and $k = 200$ N/m, calculate the natural frequency ω_n in rad/s.
(iii) Give an example of a constraint force and explain why it does no work in the direction of the constraint.
(iv) State two applications of gear trains in mechatronics systems.

Q2

CO- 2;
SO- 2;
BL- 3

Two masses $m_1 = 10$ kg and $m_2 = 6$ kg are connected by a light, inextensible string passing over a massless, frictionless pulley. The system is released from rest. Take $g = 9.81$ m/s². [3]

(a) Draw the free-body diagram for each mass and write the equation of motion for m_1 .
(b) Write the equation of motion for m_2 .
(c) Find the acceleration of the system.
(d) Calculate the tension in the string.
(e) Find the velocity of the masses after 3 seconds.
(f) Calculate the distance traveled by each mass in 3 seconds.

Q3

CO- 2;
SO- 2;
BL- 3,4

A solid sphere of mass $m = 5$ kg and radius $R = 0.2$ m rolls without slipping down an incline of angle $\theta = 30^\circ$. Take $g = 9.81$ m/s². [3]

(a) Calculate the moment of inertia of the solid sphere about its center.
(b) State the no-slip constraint relating linear and angular quantities.
(c) Write the expression for total kinetic energy (translational + rotational) of the rolling sphere.
(d) Find the linear acceleration of the sphere down the incline.
(e) Compare this acceleration to that of a frictionless sliding block on the same incline.
(f) Find the velocity of the sphere after rolling 3 m from rest.

Q4

CO- 2,3;
SO- 2,3;
BL- 4,5

A two-degree-of-freedom spring-mass system has $m_1 = m_2 = 1$ kg, $k_1 = k_3 = 100$ N/m (grounding springs), and coupling spring $k_2 = 50$ N/m connecting the two masses. [3]

(a) Write the equations of motion for mass m_1 and mass m_2 .
(b) Express the equations in matrix form $[M]\{\ddot{x}\} + [K]\{x\} = \{0\}$.
(c) Write the mass matrix $[M]$ and the stiffness matrix $[K]$.
(d) Set up the characteristic equation $\det([K] - \omega^2[M]) = 0$ and expand it.
(e) Solve for the two natural frequencies ω_1 and ω_2 .
(f) Describe the physical meaning of the two mode shapes.